Vidhata Jayaraman

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Education

Stanford University Start: Sept. 2026
Ph.D. in Electrical Engineering

University of Illinois Urbana-Champaign (UIUC) Graduated: May 2026
B.S. in Computer Engineering | B.S. in Mathematics Distinction: Highest Honors
GPA: *3.97/4.0 | 4.0/4.0* Dean's List (Top 20% of Student Body)
Thesis: "Information-theoretic Analysis of Knowledge Distillation"

Research Interests

Machine Learning Theory, High-Dimensional Statistics, Information Theory, Optimal Transport, Quantum Information

Relevant Coursework

Pure Math: Real Analysis, Linear Algebra, Abstract Algebra, Ordinary & Partial Differential Equations, Complex Analysis, Graph Theory, Probability & Measure, Algebraic Topology

Applied Math/CS: Machine Learning, Optimization, Deep Generative Models, Random Processes, Information Theory, Analog Signal Processing, Digital Signal Processing, Algorithms & Models of Computation, Quantum Information Theory, Cryptography, Optimal Control Theory

Research Experience

Research with Professor Ayfer Özgür September 2026 – Present
- *Limits of Antidistillation*: Analyzing a theoretical formulation of antidistillation specific to Transformer models to derive both impossibility results as well as positive regimes.

Senior Thesis (Co-advised by Prof. Rayadurgam Srikant and Prof. Lav Varshney) September 2025 – May 2026
- *Information-theoretic limits of Knowledge Distillation* (Primary researcher). Developed an information-theoretic framework for knowledge distillation, deriving fundamental limits on student performance with emphasis on LLMs.

Research with Professor Lav Varshney February 2024 – August 2026
- *Principled Antidistillation*: Creating a mathematical formulation of distillation attacks and deriving principled methods for antidistillation (Publications: 1, 3).
- *2-stage retrieval in Transformers*: Analyzing an Associative Memory model to explain 2-stage retrieval in Transformers. (Publications: 2, 4)
- *Visual redundancies in VLMs*: Found redundancy in visual token representations in VLMs and justified their existence rigorously (Publications: 6, 7).
- *Emergent Capabilities in Transformers*: Investigated Modern Hopfield Networks and neural associative memories to analyze emergent capabilities as model scale increases and their connection to Transformer architectures.
- *Mitigating Catastrophic Forgetting in LLMs*: Identified layer-level clustering patterns in LLMs post-continual learning and designed a switch network to mitigate catastrophic forgetting (Publication: 5).

Research with Professor Xu Chen March 2023 – February 2024
- Developed a Physics-Informed Neural Network (PINN) using the Deep Galerkin Method (DGM) to model voltage and electric field from a charged circle/sphere inside a grounded box (code available upon request).
- Work was adapted by Samsung engineers for use in internal modeling applications.
- Built an operator estimator for RLC (Resistor–Inductor–Capacitor) circuits to generalize across arbitrary parameters.

Publications & Presentations

(Citations in reverse chronological order)

1. Hartman, M.†, **Jayaraman, V.A.**†, Choraria, M., Varshney, L.R. (2026). Hiding in Plain Sight: Detectability-Aware Antidistillation of Reasoning Models *Under review for ICLR 2027*
2. Choraria, M., Gerogiannis, A., **Jayaraman, V.A.**, Mani, A., & Varshney, L.R. (2026). Context-Gated Associative Retrieval: From Theory to Transformers

3. Hartman, M.†, **Jayaraman, V.A.**†, Choraria, M., Varshney, L.R. (2026). Protecting the Trace: A Principled Black-Box Approach Against Distillation Attacks. *Presented: TAIGR Workshop @ ICML 2026*
4. Choraria, M., Gerogiannis, A., **Jayaraman, V.A.**, Mani, A., & Varshney, L.R. (2026). Towards Context-Based Retrieval in Associative Memories. *Presented: NFAM Workshop @ ICLR 2026* **[Oral Presentation]**
5. Wu, X., Hartman, M.*, **Jayaraman, V.A.***, & Varshney, L. R. (2026). Continual Instruction Tuning via Increasing Experts. *IEEE Journal of Selected Topics in Signal Processing.* doi: 10.1109/JSTSP.2026.3666304
6. Hartman, M.†, **Jayaraman, V.A.**†, Choraria, M., Bhimaraju, A., & Varshney, L.R. (2025). Skip-It? Theoretical Conditions for Layer Skipping in Vision-Language Models. Preprint arXiv:2509.25584 *Presented: ICML 2026 & CSLSC 2026*
7. Hartman, M.†, **Jayaraman, V.A.**†, Choraria, M., Bhimaraju, A., & Varshney, L. R. (2025). Unmasking the functionality of early layers in VLMs *Presented: eXCV Workshop @ ICCV 2025*
8. **Jayaraman, V.A.**, Dagommer, M. (2025). pADME Explainability and Uncertainty Quantification, Poster available upon request.
9. Bernstein, H.C., Bindel, S.R., McKibben, M.A., & **Jayaraman, V.A.** (2024). Planning Model based on Projection Methodology Bayesian Discrete Extended (PM2-BDE) *(Under internal review)*, Manuscript available upon request

† denotes joint-first author and * denotes equal contribution.

Industry Experience

Argonne National Laboratory | *Research Intern* January 2026 – May 2026
- Designed multimodal deep learning architectures to fuse high-dimensional molecular, imaging, and clinical data, enabling robust cancer biomarker discovery and optimized target population selection for precision medicine.
- Implementing Privacy-Preserving Federated Learning (PPFL) protocols to orchestrate distributed training across disparate healthcare facilities without compromising data security.

AbbVie | *Machine Learning Research Intern* May 2025 – August 2025
- Extended the SubgraphX Graph Neural Network (GNN) explainability framework to support multi-target regression and incorporate learned 3D molecular structure, overcoming limitations of the original classification-only method.
- Validated results by extracting key molecular substructures correlated with metabolic stability, directly enhancing internal site-of-metabolism analysis tools.
- Investigated Uncertainty Quantification methods, developing a novel approach utilizing metric learning (See Poster 8).

Johns Hopkins University Applied Physics Laboratory | *Data Science Intern* June 2024 – August 2024
- Implemented a Bayesian framework for Reliability Growth Planning (RGP) to model system failure intensities, enabling posterior estimation of reliability metrics during developmental testing.
- Engineered a Retrieval-Augmented Generation (RAG) pipeline to query technical documentation
- Authored a manuscript detailing the proposed Bayesian reliability methodology (Publication 9).

National Institute of Standards and Technology (NIST) | *Research Intern* June 2023 – August 2023
- Engineered a text anomaly detection pipeline utilizing Cloze probabilities from a fine-tuned GPT-2 language model.
- Validated against outlier analysis done in dimensionally-reduced embedding spaces (e.g., via PCA/t-SNE).

Brunswick i-JET Research Lab | *Autonomous Simulation Intern* January 2023 – May 2023
- Implemented Simultaneous Localization and Mapping (SLAM) within the Robot Operating System (ROS) to generate maps for autonomous navigation.
- Developed a simulation environment for an autonomous wake generator for real-time visualization and data analysis.

Teaching and Mentorship

Algorithms & Models of Computation Classroom Assistant
August 2025 – December 2025
- Assist weekly in quiz and exam grading, and hold 2-hour-long office hours weekly.
- Writing lecture notes on the connection between the WL graph isomorphism test and the expressivity of GNNs.

Skills & Expertise

Deep Learning: PyTorch, TensorFlow
Natural Language: NLTK, spaCy, LangChain
Software: Python, C/C++, Java
Web/API: Flask, Django, HTML/CSS, Javascript
DevOps: Git, Docker
Low-Level: x86 assembly, SystemVerilog
Robotics: ROS, OpenCV
Research/Math: LaTeX